LEARNER GUIDE

Angles on a line and at a point

Angles Tell Us Why · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A straight line is half a turn, so the angles that sit together on a straight line add up to **180°**. All the way round a point is a full turn, so the angles at a point add up to **360°**. When two straight lines cross, the angles opposite each other are **vertically opposite**, and they are always equal. Every step in finding an angle needs a reason: **because** …

What you are learning to notice

- However you split a straight line, the angles add up to 180°.
- However you split the turn round a point, the angles add up to 360°.
- Where two lines cross, opposite angles are equal and neighbouring angles add up to 180°.
- A reason must be a fact that is always true, not “it looks like it”.
- Some puzzles need two facts, one after the other.

Moving through it, one step at a time

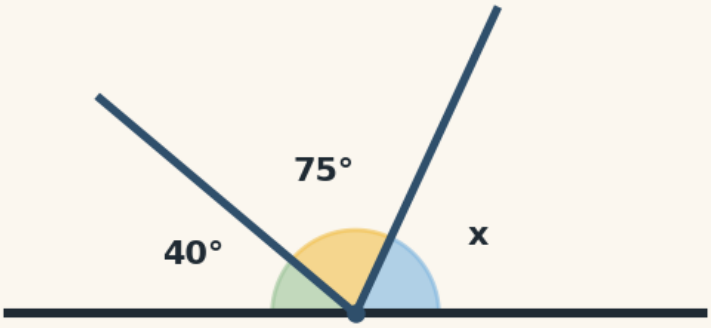
Look at the diagram: is the missing angle on a straight line, round a point, or opposite another angle where two lines cross? Add the angles you know, then take them from 180° or 360° (or copy the opposite angle). Write the answer and the reason: “x = 50°, because angles on a straight line add up to 180°.”

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — On a straight line

Find x.

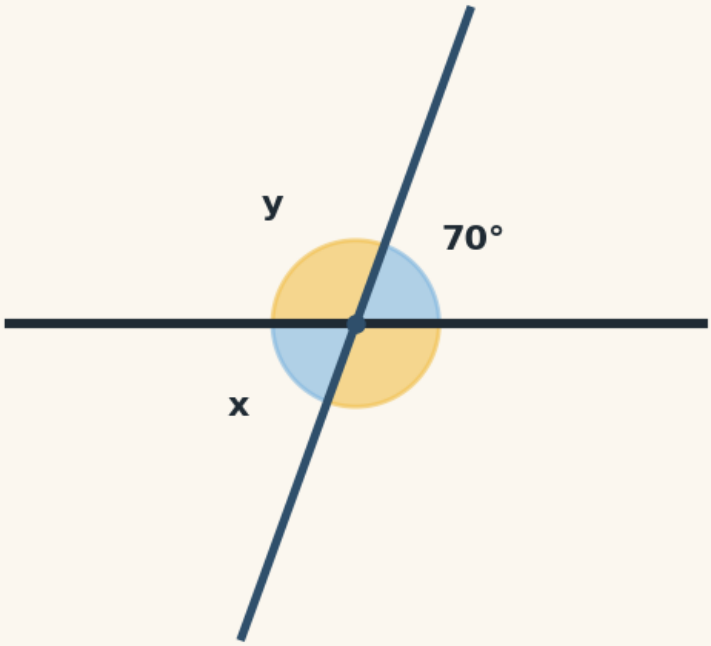


1. The three angles sit together on a straight line, so they add up to	180°
2. Add the angles you know.	75° + 40° = 115°
3. Take that from 180°.	180° – 115° = 65°

x **= 65°, because angles on a straight line add up to 180°.**

Example 2 — Where two lines cross

Find x and y.



1. x is opposite the 70° angle.	$x = 70°$
2. Reason:	vertically opposite angles are equal
3. y and 70° sit together on a straight line.	$y = 180° - 70° = 110°$
4. Reason:	angles on a straight line add up to 180°

$x = 70°$ and $y = 110°$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (half and full turns (Lesson 1)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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